

□

$$\lim_{n \rightarrow \infty} \binom{2n}{n}^{\frac{1}{n}} = L$$

A) e^2 B) 2 C) 1 D) ~~4~~

$$\begin{aligned} \binom{2n}{n}^{\frac{1}{n}} &= \left(\frac{(2n)!}{(n!)^2} \right)^{\frac{1}{n}} \\ &\approx \left(\frac{(2n)^{2n} e^{-2n}}{n^{2n} e^{-2n}} \right)^{\frac{1}{n}} \\ &= \frac{2^2 n^2}{n^2} = 4 \end{aligned}$$

Stirling approx.
(S.P.)
for large n
 $n! \approx n^n e^{-n}$

Soltz-Cesaro
Method

□

$$\begin{aligned} \lim_{n \rightarrow \infty} \left(\frac{n^{3n}}{(n!)^3} \right)^{\frac{1}{n}} &= e^3 \\ \left(\frac{n^{3n}}{(n!)^3} \right)^{\frac{1}{n}} &= \left(\frac{n^{3n}}{n^{3n} e^{-3n}} \right)^{\frac{1}{n}} \\ &= \frac{1}{e^{-3}} = e^3 \quad \checkmark \end{aligned}$$

□

$$\begin{aligned} \lim_{n \rightarrow \infty} \binom{3n}{n}^{\frac{1}{n}} &= \left(\frac{(3n)!}{n! (2n)!} \right)^{\frac{1}{n}} \\ &= \left(\frac{(3n)^{3n} e^{-3n}}{n^n e^{-n} \cdot (2n)^{2n} e^{-2n}} \right)^{\frac{1}{n}} \\ &= \frac{3^3 n^3}{n \cdot 2^2 n^2} = \frac{27}{4} \end{aligned}$$

□

let

$$k \in \mathbb{N}, \quad k > 1$$

$$\rightarrow \frac{k^k}{(k-1)^{k-1}}$$

$$\lim_{n \rightarrow \infty} \left(\frac{k^n}{n} \right)^{\frac{1}{n}}$$

$$= \left[\frac{(kn)!}{n! ((k-1)n)!} \right]^{\frac{1}{n}} \approx \left[\frac{(kn)^{kn} e^{-kn}}{n^n e^{-n} ((k-1)n)^{(k-1)n} e^{-(k-1)n}} \right]^{\frac{1}{n}}$$

$$= \frac{k^k}{(k-1)^{k-1}}$$

□

$$\lim_{n \rightarrow \infty}$$

$$\frac{\left(1 + \frac{1}{2} + \dots + \frac{1}{n-1}\right) + \frac{1}{n}}{\ln n}$$

$$= \lim_{n \rightarrow \infty} \frac{S_n}{y_n}$$

Stolz - Cesaro Method = 1

$$\frac{x_n}{y_n}$$

$y_n \rightarrow +\infty$
strictly increasing

$\{x_n\}, \{y_n\}$ are sequences that satisfy

1) $\{y_n\}$ strictly increases to $+\infty$

2)

$$\lim_{n \rightarrow \infty} \frac{x_n - x_{n-1}}{y_n - y_{n-1}} = l$$

Then

$$\lim_{n \rightarrow \infty} \frac{x_n}{y_n} = l$$

$$\lim_{n \rightarrow \infty} \frac{S_n - S_{n-1}}{y_n - y_{n-1}} = \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\ln n - \ln(n-1)}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n \ln \left(\frac{n}{n-1} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{n \ln \left(1 + \frac{1}{n-1} \right)}$$

$$= \lim_{n \rightarrow \infty} \frac{1}{\ln \left(1 + \frac{1}{n-1} \right)^n} = 1$$

□ $\binom{100}{50}$ $\binom{2n}{n}$ is always even

$$\binom{n}{r} = \frac{n}{r} \binom{n-1}{r-1}$$

□ $\binom{2n}{n} = \frac{2n}{n} \binom{2n-1}{n-1} = 2 \binom{2n-1}{n-1} \rightarrow$ always even.

□ $\binom{3n}{n}$ - divisible by 3. $\binom{nk}{n}$, $\xrightarrow{k \in \mathbb{Z}}$ div by k .

□ Let $a_n \rightarrow a$.

Find

$$\lim_{n \rightarrow \infty} \frac{1}{\ln n} \left(\frac{a_1}{1} + \frac{a_2}{2} + \dots + \frac{a_n}{n} \right)$$

A) $\ln|a|$ B) e C) 1 D) a .

□ What is the remainder when $\binom{100}{50}$ is divided by 7?

A) 0 B) 2 C) 4 D) 6

$\binom{100}{50}$ $\rightarrow$ base 7

$$\begin{array}{r} 7 \overline{) 100} \rightarrow 2 \\ 7 \overline{) 14} \rightarrow 2 \\ \underline{2} \end{array}$$

$$\begin{array}{r} 100 = (2102)_7 \\ 50 = (1101)_7 \end{array}$$

Lucas Theorem:

$$\begin{aligned} \binom{100}{50} &\equiv \binom{2}{1} \times \binom{0}{0} \times \binom{2}{1} \pmod{7} \\ &\equiv 2 \times 1 \times 2 \pmod{7} \\ &\equiv 4 \pmod{7} \end{aligned}$$

□ $\binom{100}{50}$ - what is the remainder when divided by 3?

$$100 = (10201)_3$$

$$50 = (01212)_3$$

$$\begin{array}{r} 3 \overline{) 100} \rightarrow 1 \\ 3 \overline{) 33} \rightarrow 11 \end{array}$$

$$\binom{100}{50} \equiv \binom{1}{0} \times \binom{0}{1} \times \binom{2}{2} \times \binom{0}{1} \times \binom{0}{2} \pmod{3}$$

$$= 0$$

divisible by 3.

$$\boxed{\binom{n}{r}, n, r \in \mathbb{N}, r > n}$$

$$= 0?$$

□ What is the remainder when

$$\binom{1000}{500} \text{ is divided by } 7? \rightarrow (4)$$

$$1000 = (2626)_7$$

$$500 = (1313)_7$$

$$\text{Then, } \binom{1000}{500} \equiv \binom{2}{1} \times \binom{6}{3} \times \binom{2}{1} \times \binom{6}{3} \pmod{7}$$

$$\equiv 4 \pmod{7}$$

Lucas Theorem: Let p be any prime, and

$$n = (a_k a_{k-1} \dots a_1 a_0)_p$$

$$r = (b_k b_{k-1} \dots b_1 b_0)_p$$

Then

$$\binom{n}{r} \equiv \binom{a_k}{b_k} \binom{a_{k-1}}{b_{k-1}} \dots \binom{a_1}{b_1} \binom{a_0}{b_0} \pmod{p}$$

□ $f: (0,1) \rightarrow \mathbb{R}$ be defined as

$$f(x) = \sum_{r=0}^{\infty} \binom{100+r}{r} (1-x)^r$$

Find the remainder when $f(\frac{1}{3})$ is divided by 101.

$$\square \quad \binom{n}{k} \quad \begin{array}{l} n \in \mathbb{N} \cup \{0\} \\ k \in \mathbb{N} \cup \{0\} \end{array}$$

generalize n ?

$$\binom{x}{k} \quad (x \in \mathbb{R})$$

Generalized Binomial coefficients.

$$\left(\frac{1}{2}\right)_3, \quad \binom{-2}{3}, \quad \binom{\sqrt{2}}{4}$$

$$\left. \begin{array}{l} n \in \mathbb{N} \cup \{0\} \\ k \in \mathbb{N} \cup \{0\} \end{array} \right\} \rightarrow \boxed{\binom{n}{r}} = \frac{n!}{r! (n-r)!}$$

$$= \frac{n(n-1) \dots (n-r+1) \underbrace{[(n-r) \dots 2 \cdot 1]}_{r! \text{ (left)}}}{r!}$$

$$= \frac{n(n-1) \dots (n-r+1)}{r!}$$

$$= \frac{[n]_r}{r!} \quad \text{or}$$

$[n]_r$:= falling factorial of n , of order r .

$$[5]_3 = \underbrace{5 \times 4 \times 3}$$

$$\forall x \in \mathbb{R}, \quad [x]_r = x(x-1) \dots (x-r+1)$$

ex. i) $\left[\frac{1}{2}\right]_3 = \frac{1}{2} \left(\frac{1}{2}-1\right) \left(\frac{1}{2}-2\right)$

$$= \frac{1}{2} \times \left(-\frac{1}{2}\right) \times \left(-\frac{3}{2}\right) = \frac{3}{8}$$

ii) $[\sqrt{2}]_4 = \sqrt{2} (\sqrt{2}-1) (\sqrt{2}-2) (\sqrt{2}-3)$

$$\square \quad \forall x \in \mathbb{R}, \quad \forall r \in \mathbb{N} \cup \{0\}$$

$$\binom{x}{r} := \frac{[x]_r}{r!}$$

ex: $\binom{-2}{3} = \frac{[-2]_3}{3!} = \frac{(-2) \times (-2-1) \times (-2-2)}{6}$

$$= \frac{(-2) \times (-3) \times (-4)}{6} = -4$$

$$\binom{2}{3} = \frac{[2]_3}{3!} = \frac{2 \times (2-1) \times \overbrace{(2-2)}^{\rightarrow = 0}}{3!} = 0$$

$$\binom{n}{r} = \frac{n(n-1) \dots \overbrace{0}^{\leftarrow 0}}{r!} = 0$$

$r > n$

$n, r \in \mathbb{N}$

$$\binom{3}{5} = \frac{[3]_5}{5!} = \frac{3(3-1)(3-2)\overbrace{(3-3)}^{\rightarrow 0}}{5!} = 0$$

Binomial Thm: (Lp)

$$\begin{aligned} (1+x)^n &= \sum_{r=0}^n \binom{n}{r} x^r \\ &= \sum_{r=0}^{\infty} \binom{n}{r} x^r \end{aligned}$$

Gen Binom Thm:

$$\binom{m}{r} (1+x)^m = \sum_{r=0}^{\infty} \binom{m}{r} x^r \quad (|x| < 1)$$

$m \in \mathbb{R}$

$$(1+x)^{-2} = \sum_{r=0}^{\infty} \binom{-2}{r} x^r, \quad (1+x)^{-3} = \sum_{r=0}^{\infty} \binom{-3}{r} x^r$$

$$\square \quad \binom{m}{r}, \quad m = -n \quad (n \in \mathbb{N})$$

$$\begin{aligned} \binom{m}{r} &= \binom{-n}{r} = \frac{[-n]_r}{r!} \\ &= \frac{(-n)(-n-1) \dots (-n-r+1)}{r!} \\ &= (-1)^r \frac{n(n+1) \dots (n+r-1)}{r!} \\ &= (-1)^r \frac{(r-1)! \cdot n(n+1) \dots (n+r-1)}{(r-1)! \cdot r!} \end{aligned}$$

$$= (-1)^r \binom{n+r-1}{r}$$

$$\square \quad (1+x)^{-n} = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} \underline{x^r}$$

$$\text{and} \quad \underline{(1-x)^{-n}} = \sum_{r=0}^{\infty} (-1)^r \binom{n+r-1}{r} (-x)^r = \sum_{r=0}^{\infty} \binom{n+r-1}{r} x^r$$

$$\begin{aligned} \text{So,} \quad \underline{(1-x)^{-2}} &= \sum_{r=0}^{\infty} \binom{2+r-1}{r} x^r \\ &= \sum_{r=0}^{\infty} \binom{r+1}{r} x^r = \sum_{r=0}^{\infty} \binom{r+1}{1} x^r \\ &= (1 + 2x + 3x^2 + 4x^3 + \dots) \end{aligned}$$

$$(|x| < 1)$$

$$\begin{aligned} \text{and,} \quad (1-x)^{-3} &= \sum_{r=0}^{\infty} \binom{3+r-1}{r} x^r \\ &= \sum_{r=0}^{\infty} \binom{r+2}{2} x^r \\ &= (1 + 3x + 6x^2 + \dots) \end{aligned} \quad \left(\binom{n}{r} = \binom{n}{n-r} \right)$$

$$\square \quad f: (0,1) \rightarrow \mathbb{R}$$

$\square$

$$\begin{aligned} f(x) &= \sum_{r=0}^{\infty} \binom{100+r}{r} (1-x)^r \quad \nearrow \\ &= \sum_{r=0}^{\infty} \binom{100+r}{r} y^r = \sum_{r=0}^{\infty} \binom{101+r-1}{r} y^r \\ &= (1-y)^{-101} = \frac{1}{x^{101}} \end{aligned} \quad \begin{aligned} y &= 1-x \Rightarrow 1-y = x \end{aligned}$$

$$\text{Now, } f\left(\frac{1}{3}\right) = 3^{101} \xrightarrow{\text{Term}} \text{when divided by } 101$$

$$3^{100} \equiv 1 \pmod{101} \quad (\text{FLT})$$

$$\Rightarrow 3^{101} \equiv 3 \pmod{101} \quad (\text{prime})$$

□ $k \in \mathbb{N}$

$$\sum_{r=0}^{\infty} \binom{m+r}{r} \left(\frac{k}{k+1}\right)^r = \sum_{r=0}^{\infty} \binom{(m+1)+r-1}{r} \left(\frac{k}{k+1}\right)^r = \left(1 - \frac{k}{k+1}\right)^{-(m+1)}$$

A) k^{m+1} B) $(k+1)^m$ C) $(k+1)^{m+1}$ D) k^m

$$\left. \begin{array}{l} \\ \\ \end{array} \right\} = \left(\frac{1}{k+1}\right)^{-(m+1)} = \underline{(k+1)^{m+1}}$$

□ $f: (0,1) \rightarrow \mathbb{R}$ defined as

$$f(x) := \frac{\sum_{r=0}^{\infty} \binom{99+r}{99} x^r}{\sum_{r=0}^{\infty} \binom{97+r}{97} x^r} = \frac{(1-x)^{-100}}{(1-x)^{-98}} = \frac{1}{(1-x)^2}$$

Find $f\left(\frac{1}{2}\right)$

A) 2

B) 8

C) 4

D) $\frac{1}{2}$

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